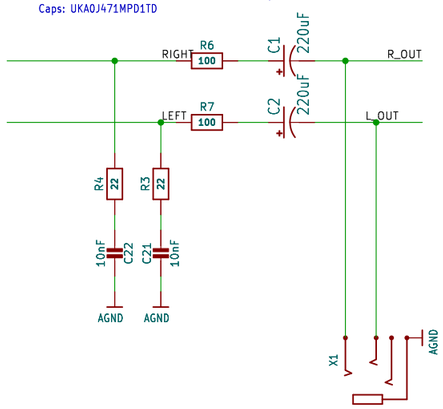
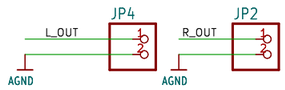
C1, C2 = 1 / (2*π*f*Xc)

16 ohm load at 20Hz cuttoff 1 / (6.2832*20hz*16) = 497µF

32 ohm load at 20Hz cuttoff 1 / (6.2832*20hz*32) = 249µF

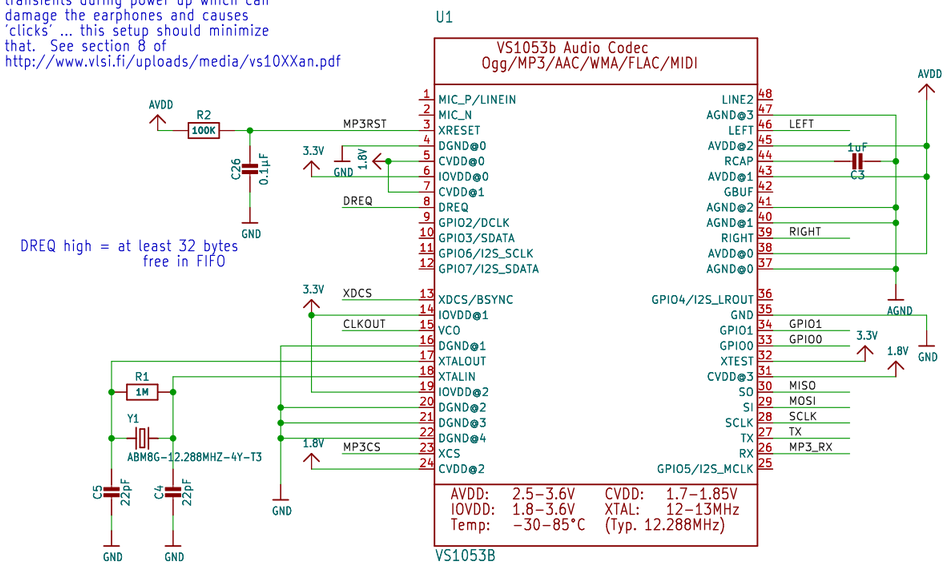
Caps: UKA0J471MPD1TD





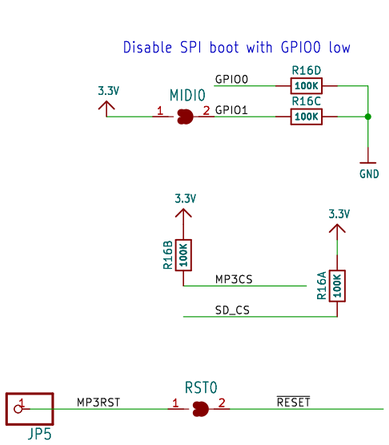
Enabling the IC too soon can cause transients during power up which can damage the earphones and causes 'clicks' ... this setup should minimize that. See section 8 of http://www.vlsi.fi/uploads/media/vs10XXan.pdf

DREQ high = at least 32 bytes free in FIFO

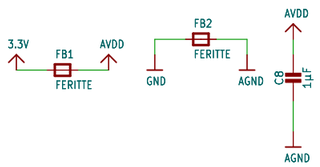


C4,C5 = 2*CL – 2*Cstray
C4,C5 = 2*10pF – 2*5pF
C4,C5 = 10pF

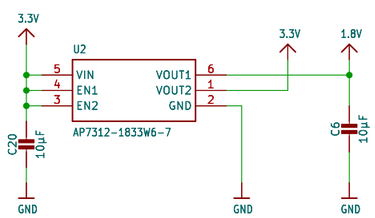
VS1053 oscillator requires 1M resistor for DC bias

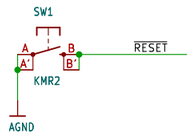


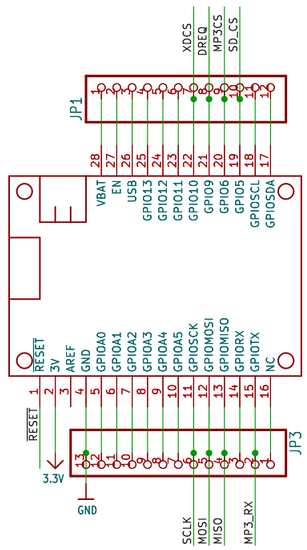
Analog Supply Filtering

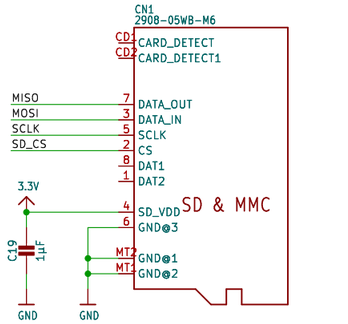


Power Supply/Decoupling









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